

[BUG] Loading hybrid quantum neuronal networks does not work with Ten

<https://github.com/PennyLaneAI/pennylane/issues/4999>

ROW 113

[BUG] Loading hybrid quantum neuronal networks does not work with TensorFlow #4999

New issue

Closed



HeyPhiS opened on Jan 5, 2024

...

Expected behavior

When using the tensorflow function `model.save` und `tf.keras.models.load_model` it is expected that the model load all layers and its ways and recreates the saved model to include all weights etc.

Actual behavior

When using `model.save` and `tf.keras.models.load_model` with a hybrid model including both classical layers and `qml.qnn.KerasLayer` the network including the quantum layer is reloaded without error, however the weights of the quantum layer are not.

This issue exists independent of the order of quantum and classical layers.

Additional information

This issue was discussed in the pennylane support thread

<https://discuss.pennylane.ai/t/error-reloading-circuit-from-qasm-string/3679>

This thread already gave rise to issue [#4856](#) to which this issue is possibly related

Source code

```
import tensorflow as tf
import pennylane as qml
import numpy as np

dev = qml.device("default.qubit")
n_qubits = 2

def create_model():
    @qml.qnode(dev)
    def circuit(inputs, weights):
        qml.AngleEmbedding(inputs, wires=range(n_qubits))
        qml.RX(weights[0], 0)
        qml.RX(weights[1], 1)
        return [qml.expval(qml.PauliZ(0)), qml.expval(qml.PauliZ(1))]

    weight_shapes = {"weights": (n_qubits,)}
    quantum_layer = qml.qnn.KerasLayer(circuit, weight_shapes, output_dim=2)

    model = tf.keras.models.Sequential()
    model.add(quantum_layer)
    model.add(tf.keras.layers.Dense(2, activation=tf.nn.softmax, input_shape=(2,)))

    model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
    return model

model = create_model()

num_points = 5
dummy_input_data = np.random.uniform(0, np.pi, size=(num_points, 2))
dummy_output_data = np.random.randint(2, size=(num_points, 2))

model.fit(dummy_input_data, dummy_output_data, epochs=1, batch_size=0)

model.save("model")

loaded_model = tf.keras.models.load_model("model")

print("")
print("saved weights:", model.layers[0].weights)
print("")
print("Loaded weights:", loaded_model.layers[0].weights)
```

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



Tracebacks

```
WARNING:tensorflow:You are casting an input of type complex128 to an incompatible dtype float32. This will decrease performance.
WARNING:tensorflow:You are casting an input of type complex128 to an incompatible dtype float32. This will decrease performance.
WARNING:tensorflow:You are casting an input of type complex128 to an incompatible dtype float32. This will decrease performance.
1/1 [=====] - 0s 349ms/step - loss: 0.5533 - accuracy: 0.8000
WARNING:tensorflow:AutoGraph could not transform <function validate_device_wires at 0x0000027642280F70> and will run it as-is.
Please report this to the TensorFlow team. When filing the bug, set the verbosity to 10 (on Linux, 'export AUTOTRANSFORM_DEBUG=1')
Cause: 'NoneType' object has no attribute '_fields'
To silence this warning, decorate the function with @tf.autograph.experimental.do_not_convert
WARNING: AutoGraph could not transform <function validate_device_wires at 0x0000027642280F70> and will run it as-is.
Please report this to the TensorFlow team. When filing the bug, set the verbosity to 10 (on Linux, 'export AUTOTRANSFORM_DEBUG=1')
Cause: 'NoneType' object has no attribute '_fields'
To silence this warning, decorate the function with @tf.autograph.experimental.do_not_convert
WARNING:tensorflow:AutoGraph could not transform <function _gcd_import at 0x000002762FE83490> and will run it as-is.
Cause: Unable to locate the source code of <function _gcd_import at 0x000002762FE83490>. Note that functions decorated with @tf.autograph.experimental.do_not_convert will not be transformed.
To silence this warning, decorate the function with @tf.autograph.experimental.do_not_convert
WARNING: AutoGraph could not transform <function _gcd_import at 0x000002762FE83490> and will run it as-is.
Cause: Unable to locate the source code of <function _gcd_import at 0x000002762FE83490>. Note that functions decorated with @tf.autograph.experimental.do_not_convert will not be transformed.
To silence this warning, decorate the function with @tf.autograph.experimental.do_not_convert
INFO:tensorflow:Assets written to: model\assets
INFO:tensorflow:Assets written to: model\assets

saved weights: [tf.Variable 'weights:0' shape=(2,) dtype=float32, numpy=array([ 0.49809802, -0.9938641 ], dtype=float32)]
Loaded weights: []
```

System information

```
Platform info:      Windows-10-10.19045-SP0
Python version:     3.10.12
Numpy version:      1.26.0
Scipy version:      1.11.2
Installed devices:
- default.gaussian (PennyLane-0.33.1)
- default.mixed (PennyLane-0.33.1)
- default.qubit (PennyLane-0.33.1)
- default.qubit.autograd (PennyLane-0.33.1)
- default.qubit.jax (PennyLane-0.33.1)
- default.qubit.legendre (PennyLane-0.33.1)
- default.qubit.tf (PennyLane-0.33.1)
- default.qubit.torch (PennyLane-0.33.1)
- default.qutrit (PennyLane-0.33.1)
- null.qubit (PennyLane-0.33.1)
- lightning.qubit (PennyLane-Lightning-0.33.1)
- strawberryfields.fock (PennyLane-SF-0.29.0)
- strawberryfields.gaussian (PennyLane-SF-0.29.0)
- strawberryfields.gbs (PennyLane-SF-0.29.0)
- strawberryfields.remote (PennyLane-SF-0.29.0)
- strawberryfields.tf (PennyLane-SF-0.29.0)
- qiskit.aer (PennyLane-qiskit-0.32.0)
- qiskit.basicaer (PennyLane-qiskit-0.32.0)
- qiskit.ibmq (PennyLane-qiskit-0.32.0)
- qiskit.ibmq.circuit_runner (PennyLane-qiskit-0.32.0)
- qiskit.ibmq.sampler (PennyLane-qiskit-0.32.0)
- qiskit.remote (PennyLane-qiskit-0.32.0)
```

The issue is also visible in the Tracebacks displayed in <https://github.com/PennyLaneAI/pennylane/issues/4856>

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



HeyPhiS added **bug** on Jan 5, 2024

timmysilv mentioned this on Jan 5, 2024

[fix keraslayer set and get attr dunder #5008](#)

```
- nois.qubit (PennyLane-0.33.1)
- lightning.qubit (PennyLane-Lightning-0.33.1)
- strawberryfields.fock (PennyLane-SF-0.29.0)
- strawberryfields.gaussian (PennyLane-SF-0.29.0)
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- strawberryfields.remote (PennyLane-SF-0.29.0)
- strawberryfields.tf (PennyLane-SF-0.29.0)
- qiskit.aer (PennyLane-qiskit-0.32.0)
- qiskit.basicaer (PennyLane-qiskit-0.32.0)
- qiskit.ibm.q (PennyLane-qiskit-0.32.0)
- qiskit.ibm.q.circuit_runner (PennyLane-qiskit-0.32.0)
- qiskit.ibm.q.sampler (PennyLane-qiskit-0.32.0)
- qiskit.remote (PennyLane-qiskit-0.32.0)
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timmysilv mentioned this on Jan 5, 2024
[fix keraslayer set and get attr dunder #5008](#)



timmysilv on Jan 5, 2024

Contributor ...

Hi **@HeyPhiS**, thanks for reporting this! I did some digging on this today, and I believe I've root-caused the issue and proposed a fix. Once it gets merged in, we'll let you know here.



timmysilv added a commit that references this issue on Jan 5, 2024

fix keraslayer set and get attr dunder ([#5008](#)) ...

Verified c89c26c



timmysilv on Jan 5, 2024

Contributor ...

the fix has been merged to a feature branch, and will be released in PennyLane v0.34, coming next week! Keep an eye out for that release, and definitely let me know if anything else comes up



HeyPhiS on Jan 8, 2024

Author ...

Great! Thank you very much very so quickly looking into this 🙌



albi3ro closed this as **completed** on Jan 9, 2024

fix keraslayer set and get attr dunder #5008

<> Code

Merged timmysilv merged 3 commits into `v0.34.0-rc0` from `keras-attr-fix` on Jan 5, 2024

Conversation 2 Commits 3 Checks 33 Files changed 3 +44 -5



timmysilv commented on Jan 5, 2024 • edited

Contributor

Context:
Fixes #4999. We re-implemented the default `setattr` and `getattr` dunder for `KerasLayer` (needed to modify the `QNode` saved on the layer) but we should have called `super()`, because `keras.layer.Layer` has its own custom implementation that we needed to use. Our `TorchLayer` already calls `super` in these cases.

Note - our existing tests were passing because the weights used had custom names ("w1", "w2", etc.), and this issue only comes up when your weights are named "weights" (which is a rather common case, I'm sure).

Description of the Change:

- Replace our default-handling of `get/set-attr` dunder with a call to `super()`
- add a test case matching the one in the bug report using the "weights" identifier

Benefits:
Models using `KerasLayer` can be saved and loaded properly!

Possible Drawbacks:
N/A

Related GitHub Issues:
Reported initially in #4999, same as #4611 but for Keras



fix keraslayer set and get attr dunder 9dd3deb

timmysilv added this to the **v0.34** milestone on Jan 5, 2024



github-actions bot commented on Jan 5, 2024

Contributor

Hello. You may have forgotten to update the changelog!
Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



Reviewers

lillian542	✓
albi3ro	✓

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

v0.34

Development

Successfully merging this pull request may close these issues.

None yet

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3 participants

